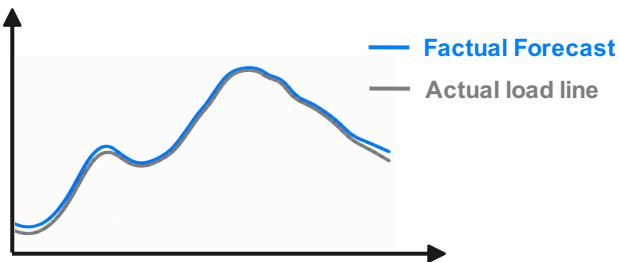


Existing News-Augmented Forecasting (Prediction-Oriented)

I know the factual forecast, but **cannot evaluate counterfactuals** under different event conditions.



Historical Load

Weather Data

News Text

Deep Predictive Model

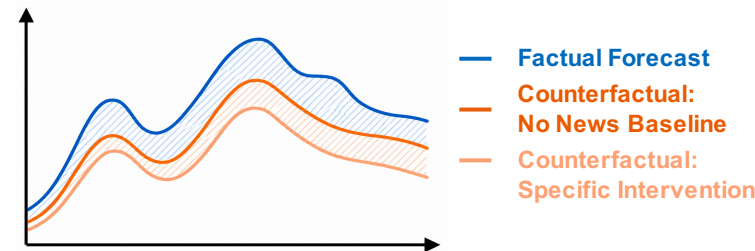
Factual Forecast

Proposed NACF Framework (Counterfactual-Oriented)

I not only know the forecast, but can also change **news interventions** to generate counterfactual predictions.



intervention



Structured Event Stream

Historical Confounders

Treatment Encoder

- ☒ Observed News
- ☐ No News Baseline
- ☐ Scenario Injection

Confounder Encoder

Representation Balancing

Varying-Coefficient Response Network